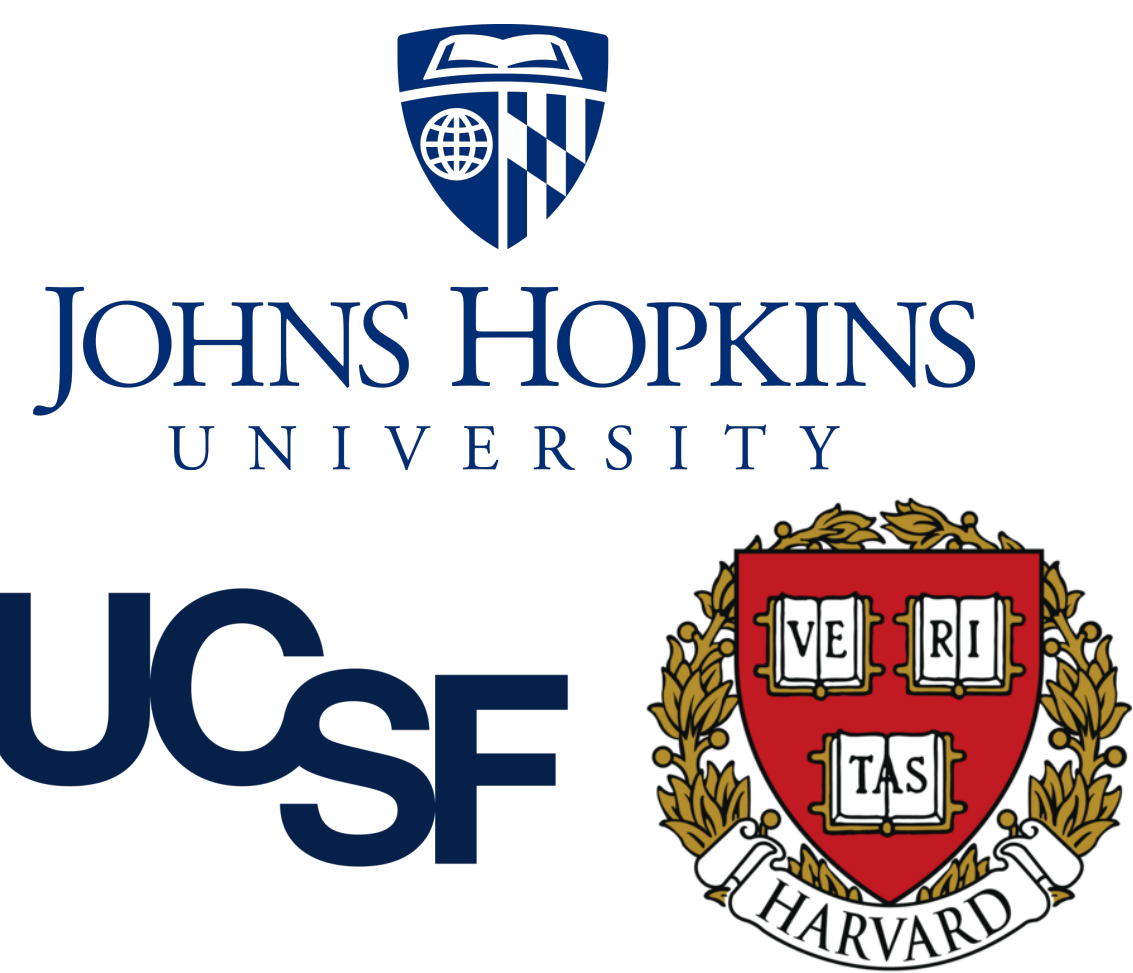




RT-Super: Learning Tumor Segmentation from Longitudinal Images and Reports

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zzhou82@jh.edu; Code (QR code): github.com/mrgiovanni/RT-Super



Learning Detection and Segmentation from Large-Scale Longitudinal Images & Reports instead of Scarce Tumor Masks

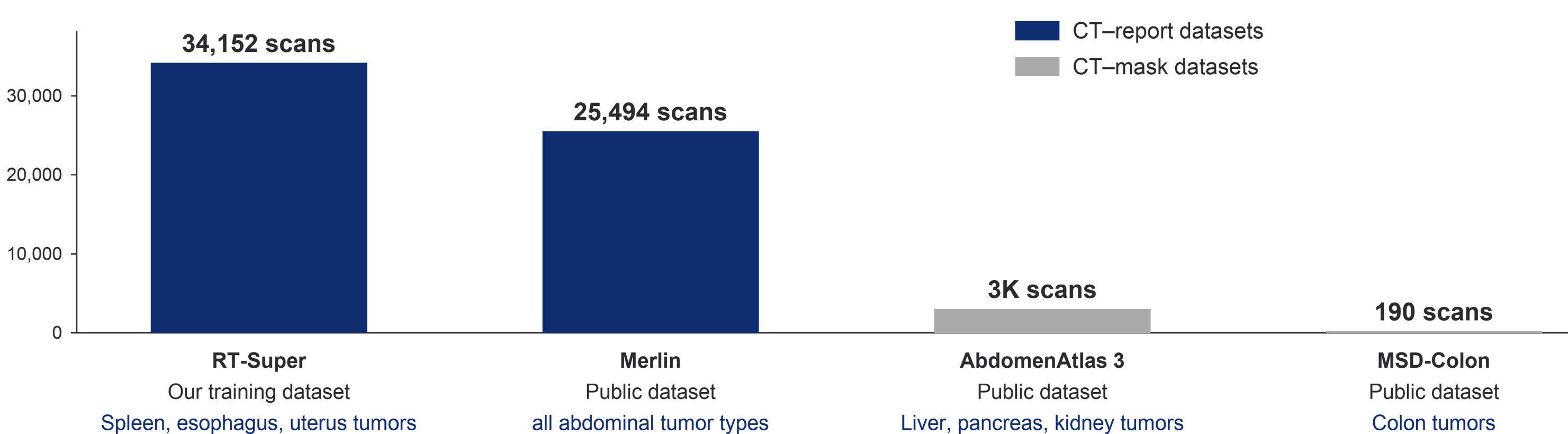
Background: Tumor segmentation can advance cancer detection and surpass VLMs and classifiers, but **segmentation datasets** are often small and lack masks for most tumor types in CT. Instead, hospitals and new public datasets have **vast, readily available data** that can guide segmentation: radiology reports, longitudinal images, and multi-phase images.

Contribution: RT-Super, an architecture that learns from large-scale longitudinal images and reports, instead of or in addition to segmentation masks. Importantly, RT-Super also improves tumor detection and segmentation when a single image and no report are available at inference – common in incidental cancer detection.

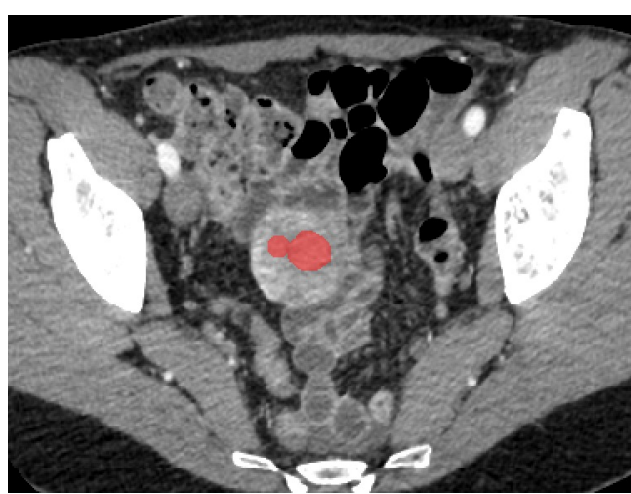
Method: A teacher network analyzes all the patient's longitudinal images and reports to create high-quality tumor masks. These masks train a student network, which sees a single image and no report. At inference, when longitudinal images and reports are unavailable, we use the student. RT-Super includes a new **CNN-Transformer architecture** and **Consistency Losses** that exploit tumor location consistency across longitudinal images.

Results: Even without training masks, RT-Super surpasses public AI models in segmenting **esophagus, uterus and spleen tumors**, which have few or no public masks. It also surpasses alternative methodologies trained on our dataset, including other methodologies that can learn from reports (e.g., CLIP and multi-task learning).

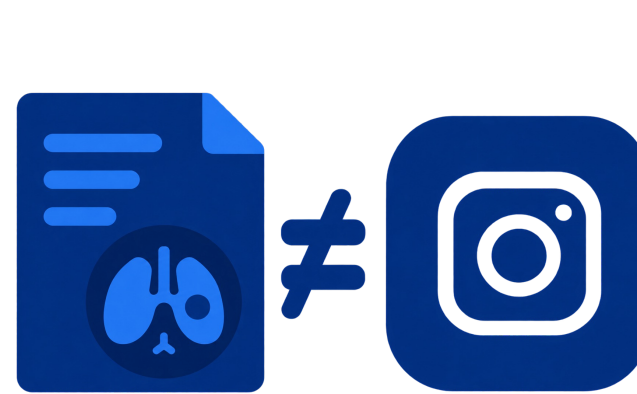
Background: Report Datasets are Much Larger than Mask Datasets & Cover All Tumor Types



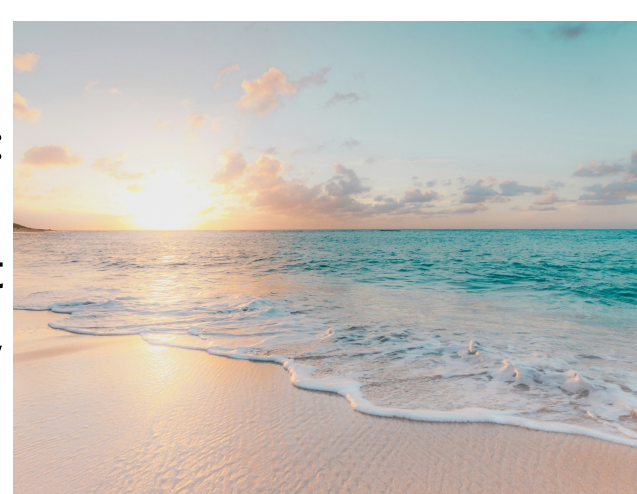
Reports ≠ Image Captions: Detailed Spatial & Lesion Information



Report: Within the upper endometrial cavity, there are two adjacent hyperattenuating lesions measuring up to 1.6-cm (series 4, image 254).



Caption: "If you want to be happy, be."

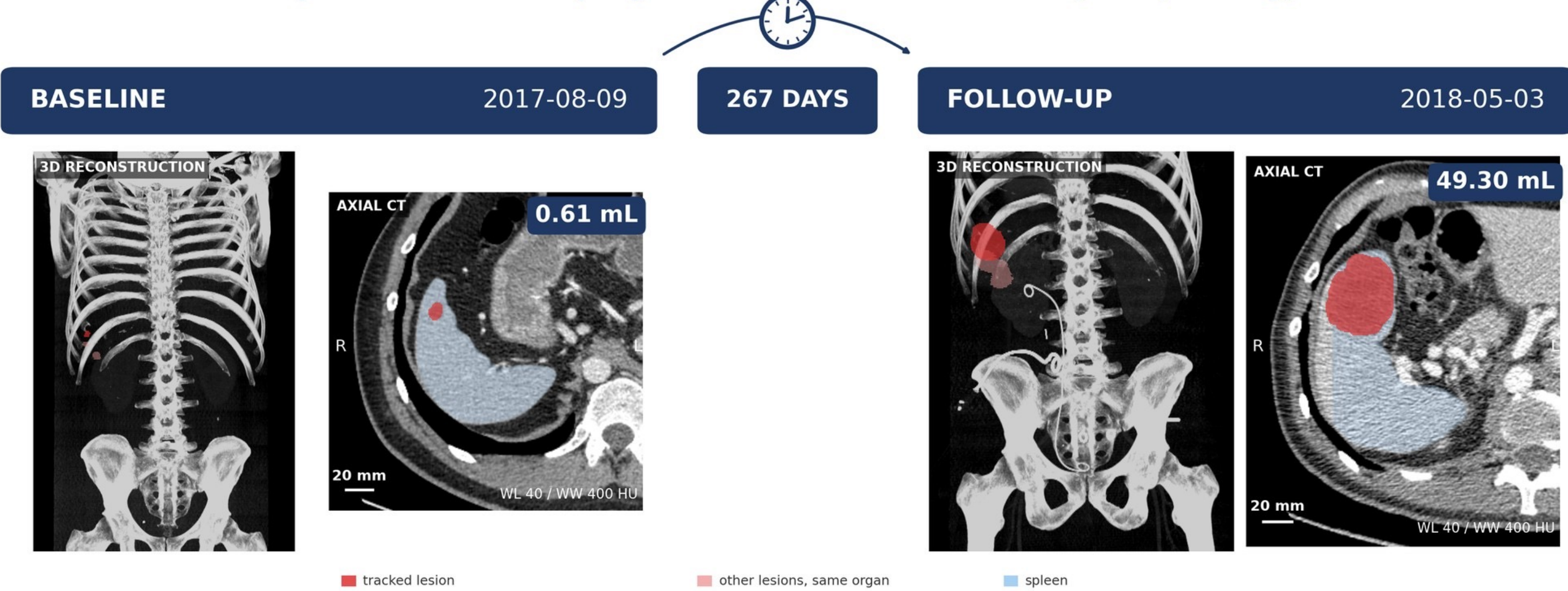


Previous work:

- ❖ MICCAI 2025: Report Supervision (R-Super), a training method to learn tumor segmentation from reports instead of masks
 - ❖ MICCAI 2025 Best Paper Award (top 2)
 - ❖ Top 1 in pancreatic tumor detection on the public PanTS dataset

To Segment or Detect Cancer, Radiologists Analyze the Patient's Previous Scans & Reports. Our AI should do the same.

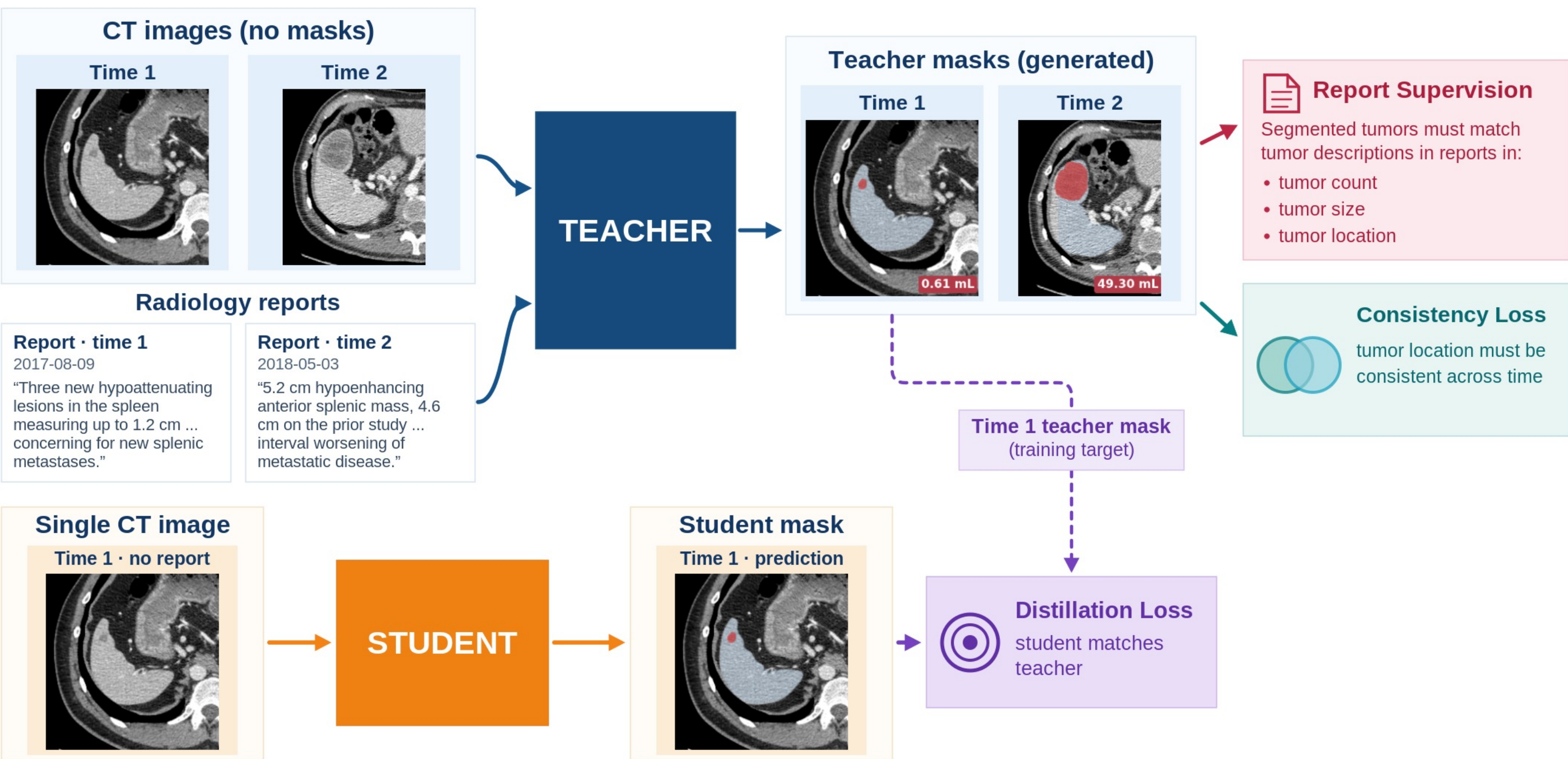
Splenic metastasis progression - patient 50637947 (male, 36-37y)



RADIOLOGY REPORT
Gallbladder: Unremarkable Spleen: Splenic granuloma.
There are 3 new hypoattenuating lesions in the spleen measuring up to 1.2 cm.
Three new splenic lesions are concerning for concerning for new splenic metastases.

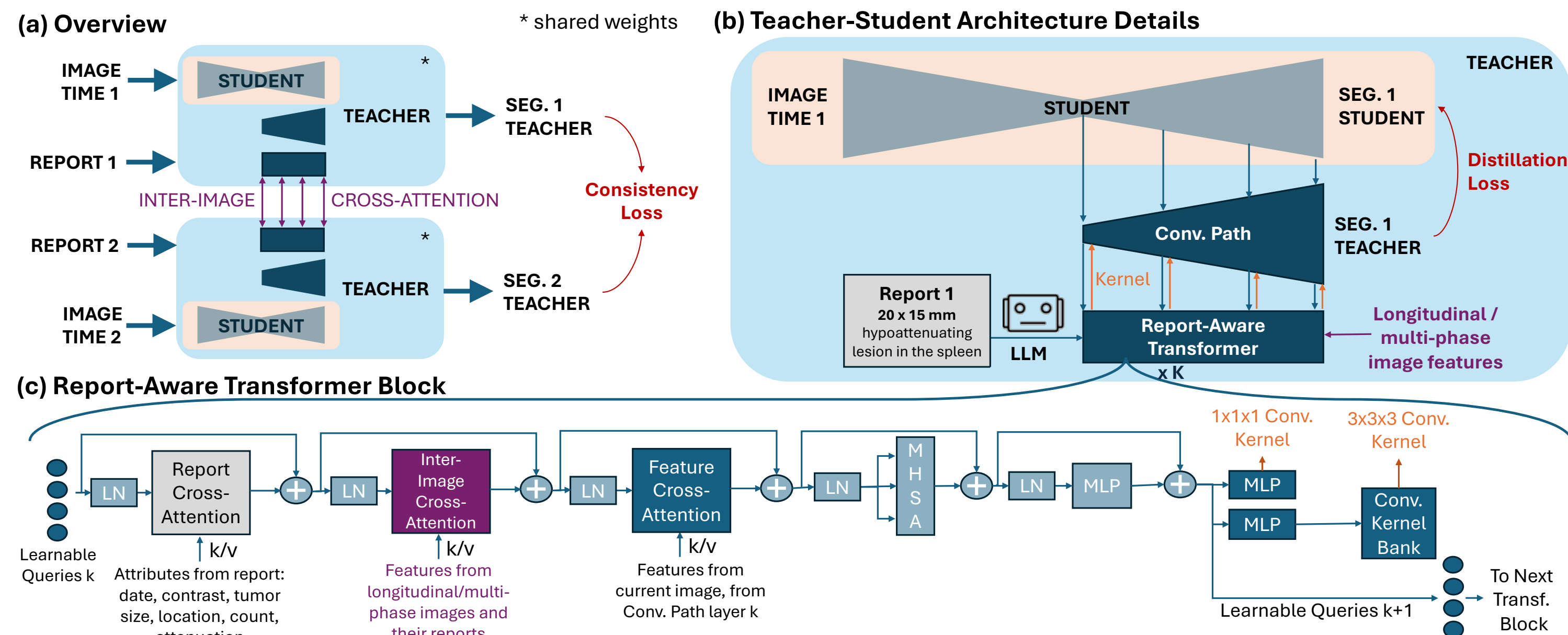
RADIOLOGY REPORT
Gallbladder: Unremarkable Spleen: 5.2 cm hypoenhancing anterior splenic mass, 4.6 cm on the prior study.
3.9 cm lobular lesion more posteriorly in the inferior spleen, previously 3.5 cm.
Interval worsening of metastatic disease, including an enlarging anterior left lower quadrant soft tissue nodule and enlarging splenic masses.

Training Method: Teacher Sees Longitudinal Images & Reports, Student Sees one Image and no Report



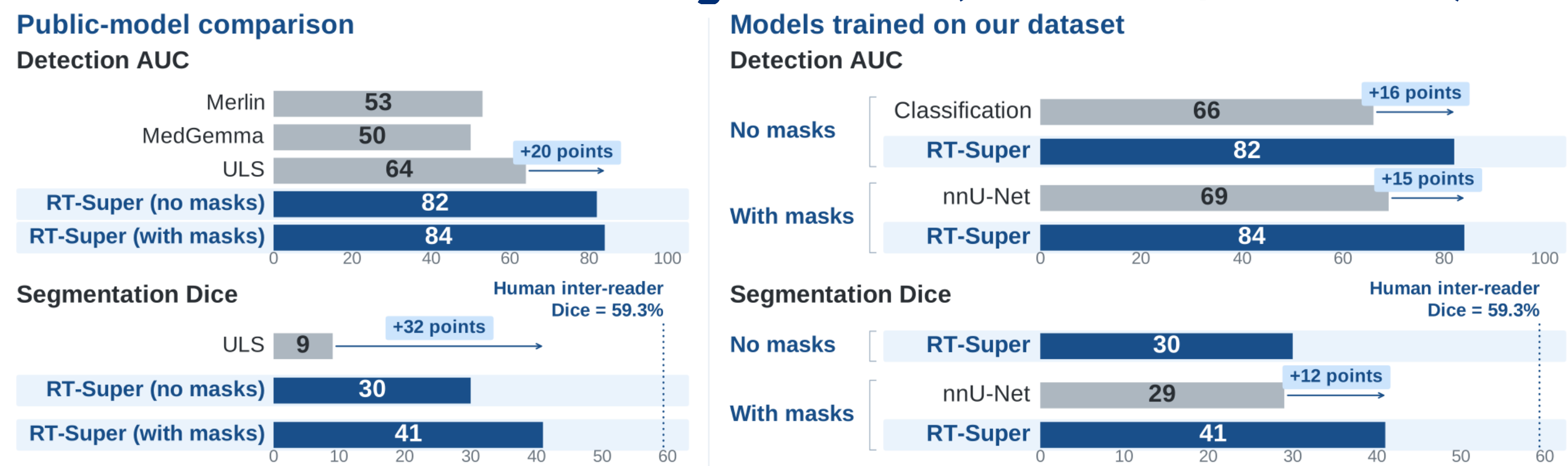
- ❖ **Inference:** use the teacher if longitudinal images are available; if not, use the student

Modular CNN-Transformer Architecture: the Student is Part of the Teacher



- ❖ **Modularity & efficiency:** the teacher uses the information from radiology reports and longitudinal images to improve the output of the student
 - ❖ The student can have any segmentation architecture
- ❖ **A transformer creates convolutions:** the teacher includes a transformer that
 - ❖ Reads information from reports and longitudinal images with cross-attention
 - ❖ Outputs convolutional kernels that process the student's outputs and features
 - ❖ Small convolutional kernels are created directly by the transformer
 - ❖ Big kernels are chosen by the transformer from a kernel bank (mixture-of-experts)

Results: RT-Super Surpasses Previous Public AI Models in multi-tumor Detection & Segmentation (+20% AUC, +32% DSC)



- ❖ **External validation & all models see one CT and no report**
- ❖ **Esophagus/spleen/uterus tumors:** few public masks, public AI struggled to detect/segment them. Learning from longitudinal CT-Reports, RT-Super fills this gap.

Internal Test: 501 CT scans, USA

model	train			spleen			esophagus			uterus			average		
	longi.	report	mask	Se	Sp	F1	Se	Sp	F1	Se	Sp	F1	Se	Sp	F1
public AI models															
Merlin [5]		x		0	100	0	0	100	0	0	100	0	0	100	0
ULS [13]			x	28	89	34	5	98	10	32	85	42	22	91	29
MedGemma [23]		x		6	95	10	0	100	0	0	100	0	2	98	3
trained on our dataset — report only															
classification [†]		x		24	78	24	78	85	84	75	82	75	59	82	61
R-Super [†] [1]		x		88	87	72	71	88	80	92	82	82	84	86	78
RT-Super [†]	x	x		71	93	73	89	80	90	87	88	84	82	87	82
trained on our dataset — report and/or mask															
CLIP [5,22]		x	x	77	81	63	82	85	87	87	80	81	82	82	77
MTL [7]		x		77	83	66	76	94	85	92	83	85	82	87	79
Models Genesis [27]		x		79	74	58	83	85	87	86	78	79	83	79	75
segmentation [12]			x	91	69	62	82	90	88	88	80	82	87	80	77
nnU-Net [16]			x	68	68	50	79	88	86	80	88	82	76	81	73
R-Super [1]		x	x	87	90	78	86	94	91	86	85	83	86	90	84
RT-Super	x	x	x	83	95	83	88	89	91	84	82	80	85	89	85

External Test: small tumors (≤ 2 cm), 418 CT scans, Turkey

model	train			spleen			esophagus			uterus			average		
	longi.	report	mask	Se	Sp	AUC DSC	Se	Sp	AUC DSC	Se	Sp	AUC DSC	Se	Sp	AUC DSC
public AI models															
Merlin [5]		x		1	99	50	-	0	100	50	-	33	87	60	-
ULS [13]			x	58	51	53	4	59	87	74	11	50	74	66	12
MedGemma [23]		x		0	100	50	-	0	99	50	-	0	100	50	-
trained on our dataset — report only															
classification [†]		x		22	85	54	-	73	72	72	-	75	56	72	-
R-Super [†] [1]		x		80	85	85	27	68	90	82	5	58	83	78	15
RT-Super [†]	x	x		67	91	81	31	83	84	86	33	30	90	79	25
trained on our dataset — report and/or mask															
nnU-Net [16]			x	60	80	71	26	62	90	78	17	25	89	58	43
R-Super [1]		x	x	73	82	81	54	89	80	90	18	83	69	80	51
RT-Super	x	x	x	66	89	82	43	89	82	91	24	75	70	78	55
ablation of RT-Super — report and mask															
no Consistency Loss	x	x	x	62	93	82	44	86	77	89	21	58	78	78	52
no inter-image cross-att	x	x	x	71	83	81	21	82	86	91	16	70	74	80	53
no dynamic kernel	x	x	x	73	91	83	47	84	87	89	24	75	72	79	48
no size-info teacher	x	x	x	69	86	80	45	71	87	86	19	100	59	80	42
no report-info teacher	x	x	x	65	87	83	37	86	85	90	18	83	70	77	52